

PETER RUN

Overview

Peter Run is a simple runner-style game for stroke patients with hemiparesis doing lower-limb rehabilitation exercises. Instead of a normal controller, the player wears a small motion sensor on their ankle called the MOVE Controller, and their prescribed leg movements act as the game's controls. Players are adult stroke patients with weakness on one side of the body, doing supervised rehab; they may tire quickly and have limited balance, so the game should never feel punishing. They stand and use a walker for support, and a therapist or caregiver is always present and can pause or stop the session at any time.

This is a Subway Surfers-style game we're basing the core concept on that game's lane-dodging, always-moving-forward format, but swapping out taps and swipes for the four leg movements below.

Controls: MOVE Controller and Game Actions

The MOVE Controller straps to the patient's weaker ankle. Inside it, a motion sensor (a BNO085 wired to an ESP32-S3) reads the angle of the leg and turns four specific movements into controller signals, the same way a joystick or button press would. The MOVE Controller can simply be treated as a standard plug-and-play gamepad sending four button presses.

Movement	Angle range	In-game action
Hip Abduction (leg swings outward)	0°–40° to 0°–45°	Character goes right
Hip Adduction (leg swings inward)	0°–20° to 0°–25°	Character goes left
Forward Step	0°–20° to 0°–30°	Character jumps
Backward Step	0°–20° to 0°–30°	Character slides

NOTE: These movements are using the right leg, when the patient's affected leg is on the left the lateral movements (left and right) are opposite.

The MOVE Controller is the real, final input for patients. As a temporary stand-in only, the developer can map the same four actions to the WASD keys on a keyboard (W/S/A/D for jump/slide/left/right) so they can build and test the game before hardware is ready. This is purely a dev convenience, not a permanent control scheme but it needs to behave the same way the real sensor will: each action should fire once, as a single discrete trigger, the same way it will when the leg crosses into the angle range listed in the table above. Building it this way means swapping WASD for the real MOVE Controller later is a straight swap, with no changes needed to the game logic underneath. Separately from these four gameplay actions, menus (the main menu, skipping the tutorial, and picking a Borg RPE rating) are navigated with a trackpad, not the leg movements.

Gameplay Loop and Progression

The character moves forward automatically, and obstacles appear randomly, each calling for one of the four actions above. Keep the pace slow and predictable with plenty of warning before each prompt. This is rehab, not a reflex test, and there should be no fail-state; a missed prompt simply doesn't score, and never ends the session early.

The game has 5 levels in total. Each level ends once the patient finishes their assigned set of exercises. A summary screen then opens a Borg RPE Scale window: the patient uses the trackpad to pick a number from 1 to 10 showing how hard that set felt (1 being very easy, 10 being very hard). Picking 1 through 6 unlocks the next level; picking 7 through 10 keeps the patient on the current level for another try. Note for the developer: this 1–10 scale is a simplified version of the clinical Borg RPE Scale for this game; the original clinical scale usually runs from 6 to 20, so no need to be thrown off if other references use different numbers.

As levels get harder, don't speed the game up, add more obstacles to dodge instead, so the patient naturally ends up repeating each action more times per level. Difficulty should scale through repetition count, not speed.

Main Menu and Tutorial

The game opens on a main menu with four options: Play, Settings, Quit, and Tutorial. Pressing Play leads into a short tutorial gif or images showing how to use the MOVE Controller. It plays for a few seconds and then continues on its own, or the patient can skip it early using the trackpad. After the tutorial (or after skipping it), there's a short delay before the actual gameplay begins.

Session Structure and Difficulty

There's no standardized rep count for stroke patients, but 10 to 15 repetitions per joint movement is a good starting point. Leveling up also follows the patient's Borg RPE rating after a set, as described above for example, 10 reps of hip abduction rated a 2 (very light effort) means the patient is ready to level up. Make rep counts configurable (10–15 default), and keep leveling up gated by that Borg RPE check rather than something the game calculates from score or speed.

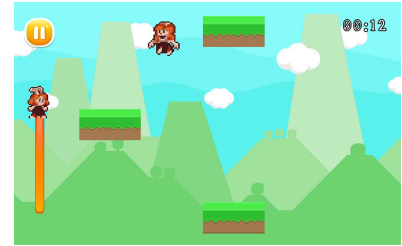
PT-Recommended Modifications and Safety

The patient always uses a walker, so the game must never require letting go of it or using both hands. A pause/stop control must always be visible and work instantly, since the therapist may need to stop play if they notice pain, fatigue, dizziness, loss of balance, abnormal movement, or shortness of breath.

Feedback and Visual Style

Visuals should be large, high-contrast, and simple, with short, encouraging feedback ("Nice step!" instead of "Missed!") and a progress bar or star rating rather than a competitive leaderboard. Match the look and feel of our earlier PETER games shown below so the games

feel like one connected family: colorful pixel-art characters and tiles, and the same kind of rounded arcade-style pause button and digital-style timer/HUD text used in the earlier titles.



Delivery Format

The finished game should be packaged as a standalone Windows .exe that doesn't require installing any separate game engine or software to run. We'll store and share it via Google Drive, so it should install or run directly on our device/console from that single file.

Proposed 3-Week Timeline

Week	Focus
Week 1	MOVE Controller / WASD input built with the correct angle-based triggers, the main menu (Play, Settings, Quit, Tutorial), and the basic lane-dodge movement and obstacle loop.
Week 2	Full gameplay loop across all 5 levels: tutorial video with skip, end-of-level summary and Borg RPE Scale gate, level unlock logic, per-level obstacle count increase, rep-count settings, and the always-visible pause/stop control.
Week 3	Visual polish to match the PETER art style, playtesting and bug fixes, then packaging into the final distributable .exe.